

Grade 7
Unit 13
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Hadley has two different ways of doing homework, on the computer or on paper. Hadley wanted to know which way of doing homework was quicker, so she asked her friends how long it took them to complete their homework for both types of assignments. Their times, in minutes, are listed below.

Online homework: {15.23, 19.12, 23.00, 45.45, 30.42, 27.32, 18.47, 36.21}

15.23 18.47 19.12 23 27.32 30.42 36.21 45.45

Paper homework:

10.34	9.55	12.56	32.48	8.21	15.44	17.52	11.01
8.21	9.55	10.34	11.01	12.56	15.44	17.52	32.48

Using the data shown above complete the table below.

	Online Homework	Paper Homework
1. Mean	26.9025	14.63875
2. Median	$\frac{23 + 27.32}{2} = 25.16$	$\frac{11.01 + 12.56}{2} = 11.785$
3. IQR	14.52	6.535
4. Range	$45.45 - 15.23 = 30.22$	$32.48 - 8.21 = 24.27$

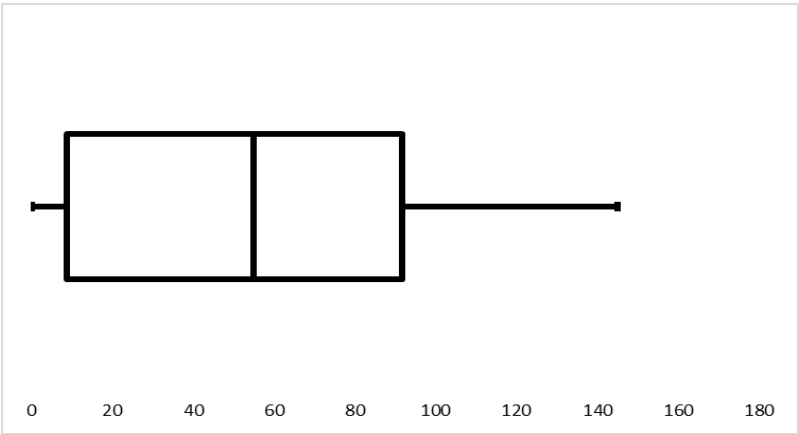
$$\frac{19.12+18.47}{2} = 18.795 \quad \frac{30.42+36.21}{2} = 33.315 \quad \frac{10.34+9.55}{2} = 9.945 \quad \frac{17.52+15.44}{2} = 16.48$$

5. What are the similarities and differences in the data? Explain your reasoning.

Differences: The mean, median, IQR, and range for online homework are larger than for paper homework.

Similarities: Same number in each data set, IQR is somewhat close

Dominic was asking his friends how long they spent playing video games, and the time they spent playing outside each week. The data for time, in minutes, spent playing video games is represented by the box plot below.



Time playing outside (in minutes): {45,10,0,32,145,100,65,75,89,5}
0, 5, 10, 32, 45, 65, 75, 89, 100, 145

Using the data above, complete the table below.

	Playing Video Games	Playing Outside
6. Mean	Cannot be determined	56.6
7. Median	≈ 57	55
8. IQR	$95 - 10 = 85$	$89 - 10 = 79$
9. Range	145	145

10. What are the similarities and differences in the data? Explain your reasoning.

Similarities: Range is the same and medians are close.
Differences: Mean cannot be determined from a box plot and the IQR is close but not the same.